Objective

We have two given design engines and a chat GUI, the goal is to build the **AI layer** that connects them.

Two product types are considered **splitter silencers** and **acoustic panels** (flat absorber walls with stratified layers), a sales assistant should be able to ask for changes in the designs from the agent.

> "set the length to 1200mm for project xx."

> "What's the NRC on the studio panel if I make the rockwool 75 mm?"

...and it changes the project's variables and shows the regenerated drawing.

Available files

engine.py # silencer engine: params in -> 3D drawing + figures out

engine_panel.py # panel engine: params in -> layered drawing + figures out

static/index.html # the chat GUI

config.txt # fill in with Gemini API key, free tier

Files to build

1. **An MCP server** that exposes the engines as tools. At minimum a caller should be able to list the available projects and change a project's variables (which regenerates its drawing). The two products have different parameters, design the tools so the right one is used for the right product.

2. **An agent backend** behind the given GUI. It connects to your MCP server as a client, discovers the tools, and lets the LLM (free Gemini API) decide which tool to call for a given instruction.

Further Considerations

1- How would you proceed if the agent is supposed to collect data from emails and detect requested changes

2- How would you proceed if the engine is Autodesk inventor instead of the python engine